

PHYS1500J - RC6

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1. Fluid Dynamics

1.1 Some Concepts

Alternative units: 1 bar = 10^5 Pa

1 psi = 1 lb/in² = 6 895 Pa

normal atmospheric pressure $P_{atm} = 1.013 \times 10^5$ Pa
 $= 14.7 \text{ lb/in}^2 = 1 \text{ atm}$

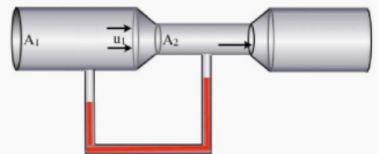
→ $P_{\text{gauge}} = P - P_{atm}$
表压, 相对大气的压力

$P = \rho gh$
 $F_{\text{buoyant}} = \rho g V_{\text{in-liquid}}$

★ $p + \frac{1}{2} \rho v^2 + \rho gy = \text{const}$ → Bernoulli's Equation.

1.2 Exercises

The wide section of a pipe (with cross-sectional area A_1) in the apparatus shown below transports an incompressible fluid of density ρ with an unknown flow speed. The wide and narrow section (of cross-sectional area A_2) of the pipe are connected via a U-shaped pipe that is filled with another incompressible fluid (marked in red) of density $\rho_f > \rho$. How can this apparatus can be used to determine the flow speed in the wider section of the tube? Give an expression for the speed in terms of A_1 , A_2 , ρ , ρ_f , the height of the fluid in the U-pipe and the acceleration due to gravity g .



① $A_1 v_1 = A_2 v_2$

② $P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$

③ $P_1 + \rho g h_1 = P_2 + \rho g h_2 + \rho g h$

static fluid,

$P = \rho g h$

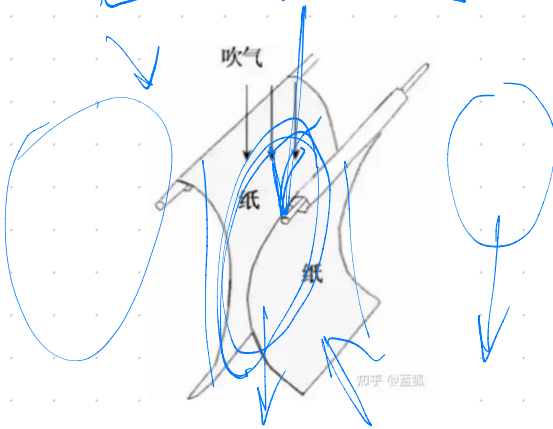
$$\begin{cases} A_1 u_1 = A_2 u_2 \\ P_1 + \frac{1}{2} \rho u_1^2 = P_2 + \frac{1}{2} \rho u_2^2 \\ P_1 - P_2 = (\rho_f - \rho) g h \end{cases}$$

$$\Rightarrow u_1 = \frac{\sqrt{2(\rho_f - \rho) g h}}{\rho \left(\left(\frac{A_1}{A_2} \right)^2 - 1 \right)}$$

$\downarrow$
 u_v

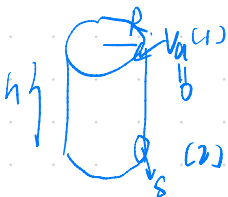
$$P + \frac{1}{2} \rho v^2 + \rho g h = C$$

$\downarrow \quad \quad \quad \uparrow$



A vertical cylindrical container has a height h and a radius R . The top is open, and the bottom has a small hole with an area S .

- (1) Find the outflow velocity of water through the small hole at the bottom.
- (2) Find the time it takes for the container to completely empty.



$$\rho g h + \frac{1}{2} \rho v_a^2 = \frac{1}{2} \rho v_0^2$$

$$v_0 \approx \sqrt{2gh}$$

$$S_A V_A = S V_0$$

$$V_A = -\frac{dh}{dt}$$

$$V_0 = \sqrt{2gh}$$

$$\Rightarrow -\frac{S}{S_A} \sqrt{2g} dh = d(\sqrt{h})$$

$$\int_0^h -\frac{S}{S_A} \sqrt{2g} dh = \int_h^0 d(\sqrt{h})$$

$$\Rightarrow t = \frac{\pi R^2}{S} \cdot \sqrt{\frac{2h}{g}}$$

$$\rho g h + \frac{1}{2} \rho v_a^2 + P = \frac{1}{2} \rho v_0^2 + P$$

$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$
 $P_{atm} \quad \quad \quad P_{atm}$

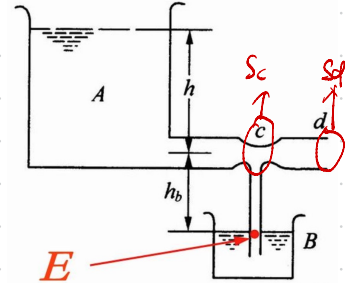
$$-\frac{S}{S_A} dh = \frac{dh}{\sqrt{2gh}}$$

$$\frac{dh}{dh} = \frac{1}{\sqrt{2gh}}$$

As shown in the figure, the lower outlet of container **A** is connected to a horizontal pipe. Point **c** in the middle of the pipe is a **constriction** (narrow section), and the **d** end of the horizontal pipe is open to the atmosphere.

(1) Find the pressure at point **c**.

(2) Under what condition can the liquid in container **B** be drawn into the horizontal pipe due to suction (i.e., when will the pipe create suction that causes inflow from B)?



(1) $v_d \approx \sqrt{2gh}$ (refer to last problem)

$$v_c = \frac{S_d}{S_c} v_d$$

$$p_0 + \frac{1}{2} \rho v_d^2 = p_c + \frac{1}{2} \rho v_c^2$$

$$\Rightarrow p_c - p_0 = \rho gh \left[1 - \left(\frac{S_d}{S_c} \right)^2 \right]$$

(2) $p_E = p_c + \rho gh_b$

$$p_c - p_0 = \rho gh \left[1 - \left(\frac{S_d}{S_c} \right)^2 \right] + \rho gh_b < 0$$

$$\Rightarrow \frac{S_d}{S_c} > \sqrt{1 + \frac{h_b}{h}}$$

2. Gravitation

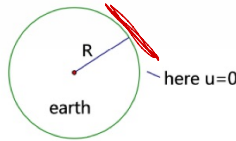
2.1 Some Concepts

$$F_G = G \frac{Mm}{r^2} = m \frac{v^2}{r}$$

a. Close to Earth's surface: $Gm = gR^2$

b. $U(r) = -G \frac{Mm}{r} + C$ $\vec{F} = -\text{grad } U = -\nabla U$
 gravitational potential energy. $\vec{F}_G$ $U = \int \vec{F}$

$$\begin{aligned}
 1 \quad U(\infty) = 0 &\Rightarrow C = 0 \text{ and } U(r) = -G \frac{Mm}{r} \\
 2 \quad U(R) = 0 &\Rightarrow -G \frac{Mm}{R} + C = 0 \Rightarrow C = G \frac{Mm}{R} \text{ and } r=R \\
 \boxed{U(r) = GMm \left(\frac{1}{R} - \frac{1}{r} \right)}
 \end{aligned}$$



Let $r = R + h$ (with $h \ll R$), then

$$\begin{aligned}
 U(r) &= GMm \left(\frac{1}{R} - \frac{1}{R+h} \right) = G \frac{Mm}{R} \left(1 - \frac{1}{1 + \frac{h}{R}} \right) \\
 &= G \frac{Mm}{R} \left(1 - 1 + \frac{h}{R} - \left(\frac{h}{R} \right)^2 + \dots \right) \quad [\text{for } h \ll R] \\
 &\approx G \frac{Mm}{R} \left(1 - 1 + \frac{h}{R} \right) = G \frac{Mm}{R^2} h = mgh
 \end{aligned}$$

逃逸速率; $\frac{1}{2} m V_{esc}^2 - G \frac{Mm}{R} = 0 \Rightarrow V_{esc} = \sqrt{\frac{2GM}{R}} \approx 11.2 \text{ km/s}$

Escape Velocity

★ Suggest to memorize

Satellites in Circular Orbit

$$v = \sqrt{\frac{GM}{r}}$$

$$T = \frac{2\pi r}{v} = 2\pi \frac{r^{\frac{3}{2}}}{\sqrt{GM}}$$

(12 points) Consider a solid sphere of water sitting in deep space. The sphere has a uniform density of $\rho = 1000 \text{ kg/m}^3$ and a radius of $R = 100 \text{ km}$.

Using what we know about gravity, find a formula for the force of gravity everywhere within the volume of the sphere. Find the pressure of the water at a distance $r = 1 \text{ meter}$ from the center of the sphere.

Shell Theorem



ball
sphere

$$M(r) = \frac{4}{3} \pi r^3 \rho$$

$$F(r) = \frac{G M(r) m}{r^2} = \frac{4}{3} \pi G \rho m r$$

$$g(r)$$

边界处 $r=R$ 处 $p=0$

$$p=0$$

$$r \rightarrow R$$

$$\frac{dp}{dr} = -\rho g(r) = -\rho \left(\frac{4}{3} \pi G \rho r \right)$$

$$\Rightarrow p(r) = \int_r^R \frac{4}{3} \pi G \rho^2 r dr = \frac{2}{3} \pi G \rho^2 (R^2 - r^2)$$

$$r=1m \quad p(1) = \frac{2}{3} \pi G \rho^2 (R^2 - 1^2)$$

$$\uparrow \uparrow \uparrow$$

$$\approx 1.4 \times 10^6 \text{ Pa}$$

I've uploaded a full version of my RC materials, and a copy of my personal notes from last year.

https://github.com/Knockingemitssparks/PHYS1500J-SJTU-Global-College---TA-Material/tree/main/RC_material